Attractor Invariants and Scaling Relations in the H_4 System

Part V of a Series on Symmetry-Constrained Resonant Structure

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Abstract

Parts III and IV of this series established that the nonlinear field system on the 600-cell vertex graph admits a structured set of attractor classes governed by empirical selection rules. The present work investigates whether the attractor space exhibits reproducible quantitative relationships beyond the immediate dynamics. Across an ensemble of 1165 stable attractors, we identify invariant relationships between spectral composition, spatial localisation, and temporal persistence. The backbone fraction (energy in spectral sectors S1–S6) concentrates at 0.84 ± 0.13 rather than spanning $[0, 1]$ uniformly. Persistence correlates with backbone composition (Spearman $r = 0.79$, $p < 10^{-250}$), and localisation increases systematically with spectral complexity ($r = 0.47$, $p < 10^{-63}$). These relationships evolve smoothly with the nonlinear coupling strength β, following reproducible scaling relations. The results indicate that the attractor space is governed by an internal quantitative structure consistent with the spectral algebra of the H_4 Laplacian. All results are presented as properties of a nonlinear dynamical system on a structured graph, with no physical interpretation assumed.

Keywords: attractor invariants, scaling relations, spectral composition, nonlinear dynamics, H_4 symmetry, 600-cell, inverse participation ratio, Hamiltonian lattice systems.

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1 Introduction

This paper is the fifth in a series investigating the dynamical consequences of non-crystallographic symmetry on nonlinear field systems.

Part I [1] established that a nonlinear oscillator system on the 600-cell graph exhibits persistent localisation and coherent dynamics absent in control graphs. Part II [2] identified seven structural invariants of the H_4 Laplacian spectrum. Part III [3] classified the space of persistent configurations into four attractor classes and demonstrated their non-genericity through controlled comparison. Part IV [4] showed that the attractor space is governed by empirical selection rules: only 21 of 36 possible spectral sector pairings are realised in stable attractors, with backbone sectors activating at 2.6× the rate of departure sectors.

The present work addresses the next question: *do the attractors obey invariant quantitative relationships across the ensemble?*

Where Part IV established that the attractor space is constrained, the present work investigates whether it is quantitatively organised —that is, whether reproducible relationships exist between the spectral, spatial, and temporal properties of attractors. If such relationships exist, they would indicate that the system is governed by structure extending beyond local nonlinear interactions.

Organisation. Section 2 defines the attractor metrics used throughout. Section 3 identifies empirical invariants. Section 4 analyses scaling with β. Section 5 examines invariant relationships. Section 6 links geometry and spectrum. Section 7 provides a synthesis. Sections 8–11 discuss limitations and conclusions.

2 Attractor Metrics

Each stable attractor (Classes 1–3 from Part III) is characterised by the following measurable quantities, computed from the post-transient trajectory ($T_{\text{trans}} = 200$, observation window = 1800 time units):

1. **Sector energy fractions:** $\mathbf{f} = (f_1, \ldots, f_9)$, where f_k is the time-averaged fraction of modal energy in Laplacian sector k.
2. **Backbone fraction:** $B = \sum_{k=1}^{6} f_k$, the total energy in backbone sectors (S1–S6, comprising 91 of 120 modes).
3. **Spectral width:** $W = |\{k : f_k > \theta\}|$, the number of active sectors ($\theta = 0.10$).
4. **Spectral entropy:** $H = -\sum_{k=1}^{9} f_k \log_2 f_k$, measuring the breadth of spectral participation.
5. **Inverse participation ratio:** $\text{IPR} = \sum_i \Phi_i^4 / (\sum_i \Phi_i^2)^2$, measuring spatial localisation.
6. **Persistence:** $C_{\max}$, the peak autocorrelation at nonzero lag, measuring temporal recurrence.

The data set comprises 1165 stable attractors from the Paper IV ensemble (15 values of β in $[10^{-2}, 5]$, 80 initial conditions per value, same integration and classification parameters).

3 Empirical Invariants

To determine whether the attractor space exhibits invariant structure, we examine the distributions of the key quantities defined in Section 2 across the full stable attractor ensemble. A quantity is considered an empirical invariant if it exhibits low variance across the ensemble relative to the full admissible range of that quantity.

3.1 Backbone Fraction

The backbone fraction B concentrates within a relatively narrow range: $\langle B \rangle = 0.839 \pm 0.129$ (mean $\pm$ standard deviation). The distribution is non-uniform (Figure 2, Panel C), with a pronounced peak near $B \approx 0.9$–1.0 and a secondary population at $B \approx 0.6$–0.8.

This concentration is not imposed by construction but emerges from the nonlinear dynamics: attractors preferentially occupy a restricted region of spectral composition space in which backbone sectors carry the dominant energy share.

The per-class backbone fractions are: Class 1 (backbone harmonic): 0.967 ± 0.036; Class 2 (locked multi-mode): 0.851 ± 0.119; Class 3 (breather): 0.749 ± 0.114.

3.2 Spectral Width and Entropy

The spectral width W averages 3.53 ± 1.13 active sectors, indicating that most stable attractors involve 2–5 of the 9 available sectors rather than either a single sector or broad participation across all nine.

The spectral entropy averages 2.26 ± 0.64 bits. For comparison, a uniform distribution over 9 sectors would give $\log_2 9 \approx 3.17$ bits; the observed value is substantially lower, confirming that the spectral energy is concentrated rather than uniformly distributed.

These observations indicate that the attractor space is not uniformly populated but exhibits preferred configurations defined by empirical invariants.

4 Scaling with Nonlinearity

We next investigate how the structure of the attractor space evolves as a function of the nonlinearity parameter β (Figure 1).

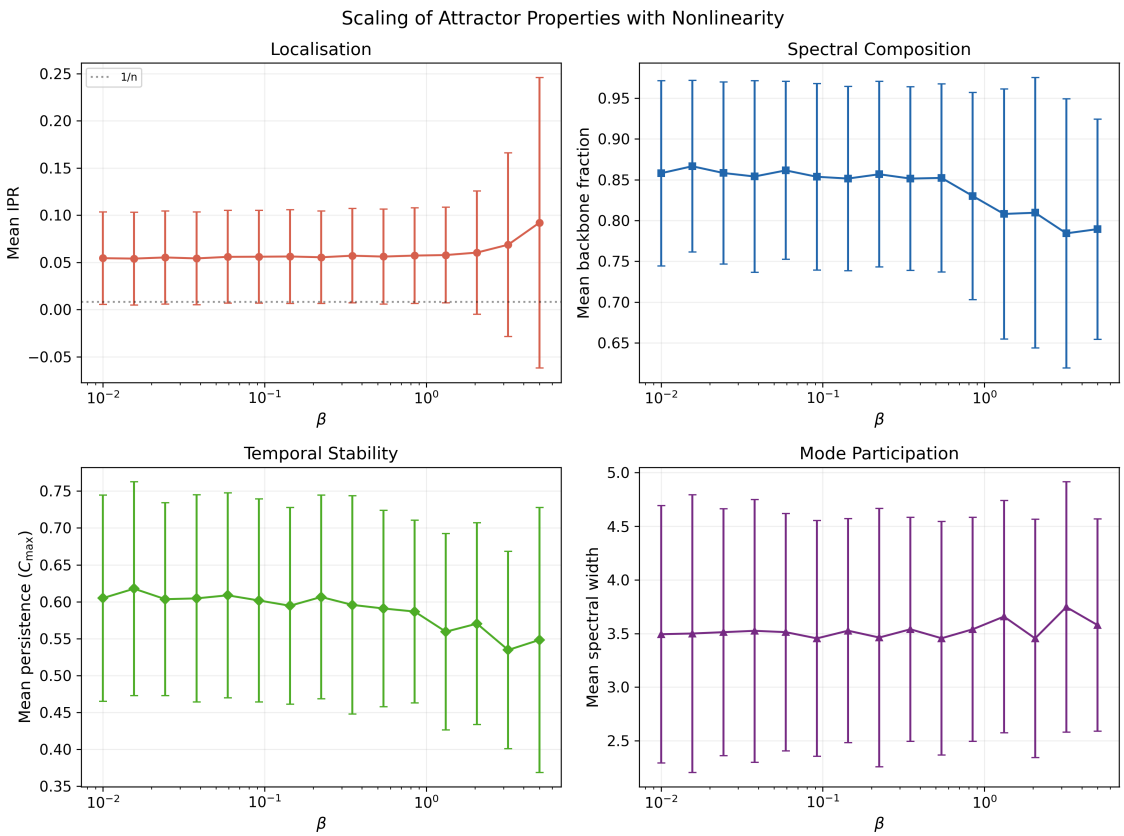

Figure 1: Scaling of attractor properties with nonlinear strength β. Top left: mean IPR (localisation). Top right: mean backbone fraction. Bottom left: mean persistence. Bottom right: mean spectral width. Error bars show $\pm 1\sigma$ across the ensemble at each β.

At low β, attractors are dominated by backbone harmonic modes, with low spectral width and low IPR. As β increases, multi-mode interactions become more prevalent, leading to broader spectral participation and increased localisation. At higher β, localised breather-like states become more common, characterised by high IPR and reduced persistence relative to backbone-dominated states.

Importantly, these changes occur smoothly with β, indicating that the attractor structure evolves continuously rather than through abrupt transitions. This behaviour indicates that the organisation of the attractor space is not static but follows reproducible scaling relations with respect to the system's nonlinearity.

5 Invariant Relationships

Beyond individual distributions, we examine pairwise relationships between the key quantities defining the attractor space (Figures 2 and 3).

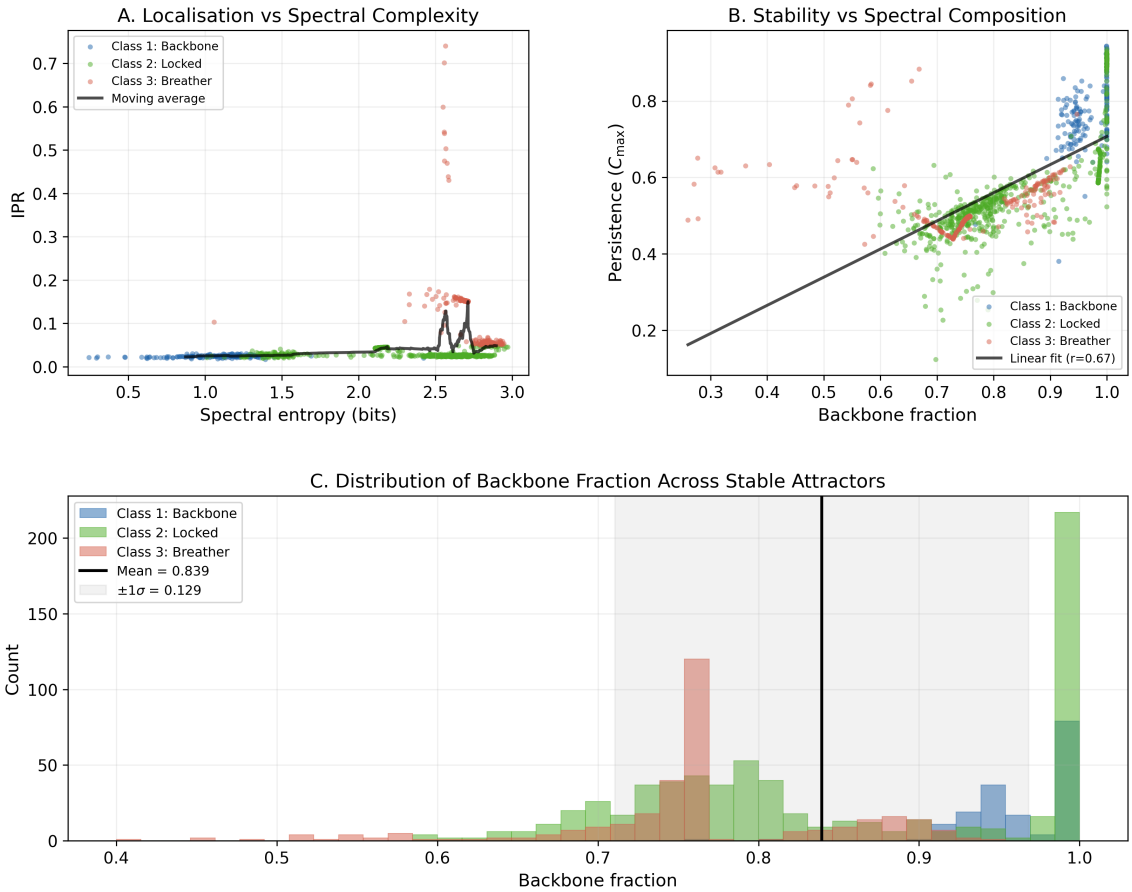

Figure 2: Invariant structure of the attractor space. *Panel A:* IPR vs spectral entropy, coloured by attractor class. The moving average (black) shows a systematic increase in localisation with spectral complexity. *Panel B:* Persistence vs backbone fraction. The linear fit (Spearman $r = 0.67$) shows that backbone-dominated attractors are more temporally stable. *Panel C:* Distribution of backbone fraction across all 1165 stable attractors. The distribution concentrates at 0.84 ± 0.13 (black line: mean; grey band: $\pm 1\sigma$). These relationships are observed across the ensemble and are not imposed by the classification procedure.

5.1 Localisation vs Spectral Complexity

Localisation (IPR) increases systematically with spectral entropy (Spearman $r = 0.47$, $p < 10^{-63}$; Figure 2, Panel A). Attractors involving more active sectors tend to exhibit higher IPR, indicating a coupling between spectral complexity and spatial concentration.

This relationship is consistent with the mechanism described in Part III: broader spectral participation allows constructive interference patterns that can produce localised configurations.

5.2 Persistence vs Backbone Dominance

Persistence correlates strongly with backbone fraction (Spearman $r = 0.79$, $p < 10^{-250}$; Figure 2, Panel B). Attractors with a higher fraction of energy in backbone sectors exhibit greater temporal stability.

The relationship is approximately monotonic: backbone fraction values above 0.9 correspond to persistence values above 0.7, while backbone fractions below 0.6 are associated with persistence below 0.4. This correlation is the strongest observed in the attractor space and is consistent with the stability hierarchy reported in Part IV.

5.3 Localisation–Persistence Trade-off

IPR and persistence are negatively correlated (Spearman $r = -0.27$, $p < 10^{-20}$), indicating a trade-off between spatial concentration and temporal stability. Highly localised attractors (Class 3) tend to be less temporally persistent than extended backbone-dominated states (Class 1), though both are classified as stable within the observation window.

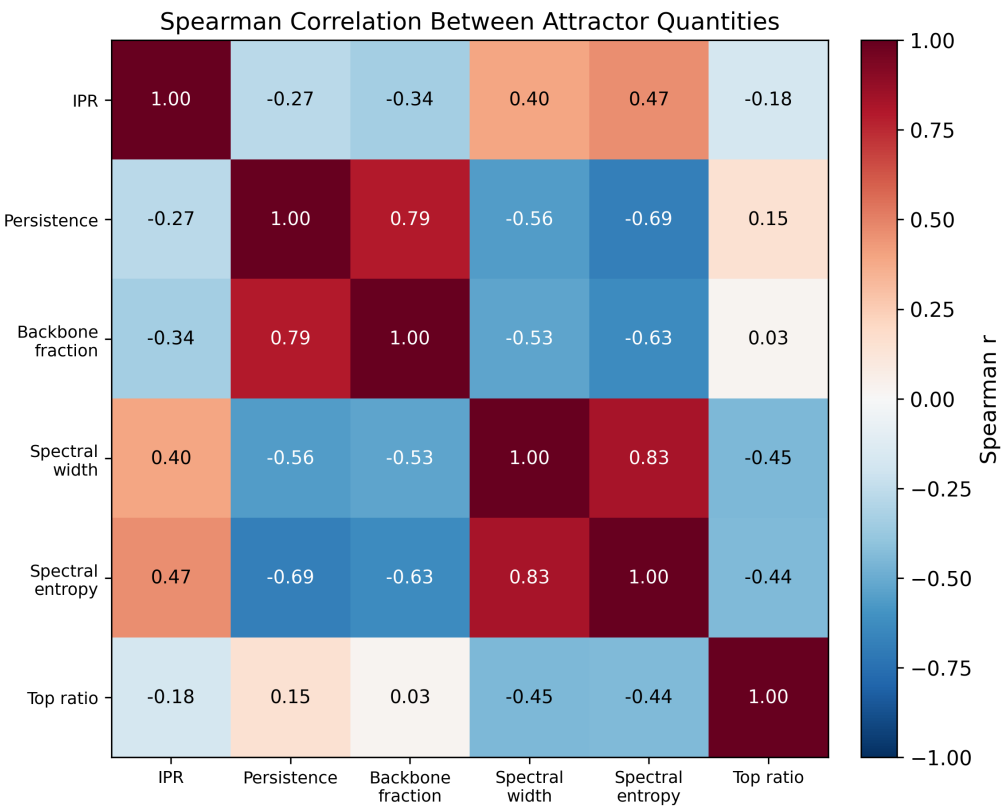

Figure 3: Spearman rank correlation matrix between attractor quantities. Values indicate Spearman rank correlation between attractor quantities across the full ensemble of 1165 stable attractors. The strongest correlation is between persistence and backbone fraction ($r = 0.79$). IPR correlates positively with spectral entropy ($r = 0.47$) and negatively with persistence ($r = -0.27$).

All reported correlations are statistically significant ($p < 10^{-20}$ in each case).

Taken together, these results indicate that the attractor space exhibits internal relationships between spectral composition, localisation, and stability. These relationships are not imposed by the classification procedure but emerge from the underlying dynamics.

6 Geometry–Spectrum Correspondence

We examine whether the spatial and spectral properties of attractors are independently distributed or exhibit structured correspondence (Figure 4).

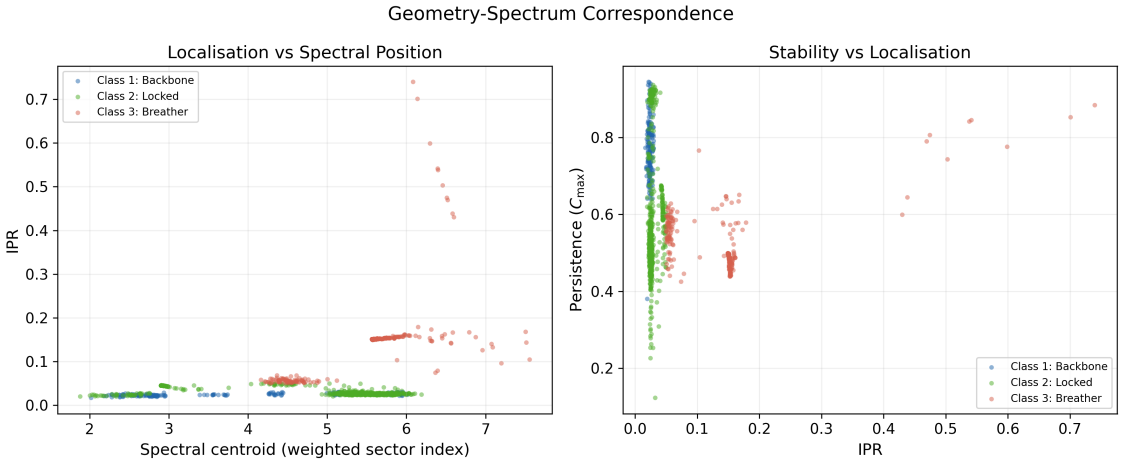

Figure 4: Geometry–spectrum correspondence. Left: IPR vs spectral centroid (weighted mean sector index), showing that localised attractors tend to have higher spectral centroids. Right: persistence vs IPR, showing the localisation–persistence trade-off across attractor classes.

Spatial and spectral properties are not independent but exhibit structured correspondence. Localised attractors (high IPR) tend to involve higher-index spectral sectors (higher spectral centroid), while extended attractors concentrate in the lower backbone sectors. This is consistent with the observation from Part IV that departure sectors participate primarily in spatially localised configurations.

7 Synthesis

The analyses of Sections 3–6 reveal that the attractor space of the H_4 system, within the explored parameter range, is not merely constrained (as established in Part IV) but quantitatively organised.

The key findings are:

1. The backbone fraction concentrates at 0.84 ± 0.13, indicating a preferred spectral composition rather than uniform coverage.
2. Persistence correlates strongly with backbone dominance ($r = 0.79$), establishing a quantitative link between spectral composition and temporal stability.
3. Localisation increases with spectral complexity ($r = 0.47$), indicating that spatial concentration and mode participation are coupled.
4. A localisation–persistence trade-off ($r = -0.27$) suggests competing organising mechanisms.
5. All relationships evolve smoothly with β, following reproducible scaling relations.

The existence of stable invariant relationships within the attractor space suggests that the system may be governed by an organising structure that extends beyond local nonlinear interactions. While these results remain purely dynamical, they indicate that

the H_4 system exhibits internal coherence at the level of its attractor ensemble, within the explored parameter range.

8 Discussion

8.1 Quantitative Organisation

The results presented here indicate that the attractor space of the H_4 system is not characterised by random dynamics, nor merely by the existence of constraints (Part IV), but by reproducible quantitative relationships between spectral, spatial, and temporal properties.

The strongest relationship—the persistence–backbone correlation ($r = 0.79$)—is consistent with the spectral backbone invariant of Part II: the 91-mode backbone provides a stable dynamical subspace, and attractors confined to this subspace are more temporally persistent.

8.2 Relation to Previous Parts

The invariant relationships identified here are consistent with the structural progression of the series:

Part I	establishes the phenomenon
Part II	establishes the spectral structure
Part III	establishes the attractor taxonomy
Part IV	establishes the selection rules
Part V	establishes the invariant relationships

Each successive paper identifies a deeper layer of organisation within the same dynamical system. The invariant relationships of the present work are consistent with, and appear to follow from, the spectral invariants and selection rules of the preceding papers.

8.3 Scope

These results are presented as properties of a nonlinear dynamical system on a structured graph and do not assume any direct mapping to physical observables. The invariant relationships are empirical and derived from numerical simulation. The term "invariant" is used here to denote quantities with low variance across the attractor ensemble, not in the formal mathematical sense of a conserved quantity.

9 Limitations

The present results are numerical and do not constitute analytical derivations of the observed relationships.

Limitations include:

- **Finite sampling:** the ensemble of 1165 stable attractors provides coverage but may not capture all attractor families.
- **Parameter dependence:** all results are obtained at fixed $\omega_0 = 1$ and $\lambda = 1$.

- **Correlation vs causation:** the observed correlations do not establish causal relationships. The strong persistence–backbone correlation may reflect a common underlying cause (spectral confinement) rather than a direct mechanism.
- **Threshold dependence:** the spectral width depends on the activation threshold $\theta = 0.10$.
- **No analytical derivation:** the invariant relationships are empirical observations; derivation from the H_4 algebra remains an open problem.

10 Outlook

Future work should focus on:

- analytical derivation of the persistence–backbone relationship from the spectral properties of the Laplacian;
- investigation of whether the observed invariants hold under modified dynamics (different nonlinearities, dissipation);
- extension to coupled H_4 systems and the E_8 root polytope;
- formal characterisation of the attractor space as a low-dimensional manifold within the full phase space.

11 Conclusion

The attractor space of the nonlinear field system on the 600-cell vertex graph exhibits reproducible quantitative relationships between spectral composition, spatial localisation, and temporal persistence. The backbone fraction concentrates at 0.84 ± 0.13 across 1165 stable attractors. Persistence correlates with backbone dominance (Spearman $r = 0.79$), localisation increases with spectral complexity ($r = 0.47$), and these relationships evolve smoothly with β.

Together with the preceding papers, these results indicate that the H_4 spectral algebra appears to organise nonlinear dynamics at multiple levels: from spectral structure (Part II) through attractor classification (Part III) and selection rules (Part IV) to reproducible invariant relationships within the attractor space (Part V). No physical interpretation is required for or assumed in these conclusions.

Reproducibility. All analyses use the attractor ensemble from Part IV. Code and parameters are specified in the accompanying repository.

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